Generating Wikipedia Articles from Citations

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March 12, 2026

1 Problem Description

Wikipedia articles are usually written as concise, neutral, and well-structured summaries of knowledge that is grounded in external sources. In practice, the references of a Wikipedia article often point to books, news articles, reports, scientific papers, or official websites that support individual claims in the text. This project asks the question: *Can we reconstruct a plausible Wikipedia-style article if we are only given the cited source material, but not the original article itself?*

2 Dataset Description

We will build the dataset from existing Wikipedia articles and their references.

- **Dataset source:** Wikipedia articles together with their cited external references.
- **Type of text:** encyclopedic article text, citation metadata, and source documents such as news articles, reports, official websites, and scientific abstracts.
- **Approximate size:** about 420 Wikipedia articles.
- **Language:** English.
- **Public or collected:** collected from public sources.

To keep the task feasible and the source coverage reliable, we will only include Wikipedia articles that satisfy two conditions:

- at least 20 citations in total,
- at least 90% of these citations are accessible to us, for example not paywalled or missing.

Each sample consists of one Wikipedia article and its citation set. The original article text is used only as a reference target for evaluation, not as input to the generation system.

This is mainly a generation task, so the dataset does not contain class labels in the standard classification sense. Instead, each sample has:

- input: cited source documents and citation metadata,
- target: the original Wikipedia article text.

The dataset will be divided into training, validation, and test sets, resulting in about 300 training articles, 60 validation articles, and 60 test articles.

Preprocessing steps will include source collection, boilerplate removal, duplicate detection, passage segmentation, and filtering of inaccessible or low-quality references.

3 Methodology Approach

We plan to compare at least two approaches.

Preprocessing steps:

- extract citations from selected Wikipedia articles,
- download or collect accessible referenced documents,
- remove web noise such as navigation text or advertisements,
- split sources into passages,
- extract metadata such as title, date, publisher, and named entities.

Embedding model: We will use a transformer-based embedding model to represent source passages and support retrieval of relevant evidence for generation.

Approach 1: Baseline. A simpler baseline will rank important passages from the cited sources and create a multi-document summary, for example using extractive summarization or a lightweight generation pipeline.

Approach 2: Advanced model. A more advanced approach will use retrieval-augmented generation with a transformer model. Relevant passages will be retrieved for each article section or generation step, and the model will generate a Wikipedia-style article grounded in these passages.

If time permits, we may add a structured variant in which the system first predicts an outline or section plan and then generates the final text section by section.

4 Evaluation Metrics

The models will be evaluated automatically against the original hidden Wikipedia article.

We plan to use:

- ROUGE for lexical overlap,
- semantic similarity based on sentence or document embeddings,
- factual support checks against the cited sources,
- and structural measures such as section heading similarity or lead quality.

The dataset will be divided into training, validation, and test sets, resulting in about 300 training articles, 60 validation articles, and 60 test articles. The validation set will be used for model selection and parameter tuning, while the test set will be kept untouched for the final evaluation.

Since exact wording may differ even in a good reconstruction, we will not rely on a single metric, but combine overlap, semantic similarity, and source-grounded factuality.

5 Expected Results

We expect the transformer-based retrieval-augmented approach to outperform the baseline because it can better capture context across multiple sources and generate more coherent text.

We also expect article quality to depend strongly on the citation set. Articles with many accessible and focused references should be easier to reconstruct than broad topics with sparse or noisy citations. In general, we expect the generated articles to recover the main facts and overall topic, but to differ from the original Wikipedia article in wording, structure, and level of detail.